

Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam
(An Autonomous Institution, Affiliated to Anna University, Chennai)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Date: 07-07-2026

Course Code	UCS3501	Course Name	Computer Networks						
Course Type	Theory	Course Category	Professional Core (PC)	L	T	P	TW&SL	TH	C
				45	0	0	45	90	3
Regulation	2024		Academic Year & Batch	2026 - 27 & 2024 - 28					
Degree and Branch			B.E. Computer Science & Engineering						
Semester			V	Section: A, B & C					
Department Offering the Course			Computer Science and Engineering						
Assignment – 1			Due Date:18-08-2026						

Assignment:1 – Domain Name System (DNS) (K6, CO1, CO6, 1.4.1, 4.1.3, 4.2.1, 5.1.1, 5.2.2, 7.1.1, 8.3.1, 9.1.1, 11.2.1, 12.4.1, 13.4.1) (WP1, WP2, WP3) (SDG 9.4 & 12.5) (WK3, WK4, WK5, WK8, EA5)

Instructions: Form a team of 2 students and choose any one of the following problems. Submit the answers in A4 sheet papers for manual submission. Scan the same and submit the scanned copy as a single pdf in the LMS.

1. A cloud service provider is planning to improve the reliability, performance, and security of its Internet services by auditing the DNS configurations of its domains. As a network engineer, you are required to **design and develop** a DNS analysis framework that retrieves, classifies, and interprets DNS records from domains belonging to different hierarchical levels. Based on the observations, propose suitable recommendations to improve the DNS infrastructure.

Tasks:

i. Design and implement a methodology using the **dig** command (or an equivalent DNS utility) to retrieve DNS records for domains such as www.iit-m.ac.in, iitm.ac.in, ac.in, and in. Organize the collected records in a structured format suitable for analysis. **(15 Marks)**

ii. Develop a classification of the retrieved DNS records (**A, AAAA, NS, MX, CNAME, SOA, TXT**, etc.), **justify** the role of each record in DNS resolution, and identify any missing or misconfigured records that may affect domain availability, performance, or security. **(15 Marks)**

iii. Create a comprehensive technical report that summarizes the DNS hierarchy, observations, identified issues, and **recommends** suitable improvements to enhance the

reliability, performance, and security of the DNS infrastructure. Justify the recommendations using the collected evidence. **(10 Marks)**

2. Design and Develop a DNS Topology Mapping Framework to Visualize the Hierarchical DNS Infrastructure (K6, CO6, WK3, WK4, WK5, WK8, WP1, WP2, WP3, EA5, SDG 9.4 & 12.5)

A network operations team is tasked with documenting the DNS infrastructure of multiple organizations to understand the hierarchical arrangement of DNS servers and improve domain resolution efficiency. As a network engineer, **design and develop** a DNS topology mapping framework that automatically discovers and visualizes the DNS hierarchy for selected domains.

Tasks

- i. Design and implement** a methodology using the **dig +trace** command (or an equivalent DNS utility) to discover the DNS resolution path for domains such as www.iitm.ac.in, www.iitmandi.ac.in, and www.nic.in. Organize the collected information systematically. **(15 Marks)**
- ii. Develop** a hierarchical topology diagram illustrating the DNS resolution process, including the **Root, TLD, and Authoritative DNS servers**, along with their server names, IP addresses, and server roles. **(15 Marks)**
- iii. Create** a technical report comparing the DNS topologies of the selected domains and **recommend** suitable improvements for enhancing DNS reliability, scalability, and fault tolerance. **(10 Marks)**

3. Design and Develop a Comparative Framework to Evaluate the Impact of Geographic Distribution of DNS Servers on Network Performance (K6, CO6, WK3, WK4, WK5, WK8, WP1, WP2, WP3, EA5, SDG 9.4 & 12.5)

A multinational cloud service provider intends to optimize user experience by evaluating how the geographic placement of DNS servers influences domain resolution performance. As a network engineer, **design and develop** a framework to compare DNS infrastructures deployed across different countries.

Tasks

- i. Design and conduct** a comparative study of DNS infrastructures for domains hosted in different countries, such as www.nus.edu.sg, www.ntu.edu.sg, www.iitm.ac.in, and www.nic.in, using suitable DNS utilities. **(15 Marks)**
- ii. Develop** a comparative analysis based on parameters such as **latency, DNS resolution time, server hierarchy, number of authoritative servers, and geographic distribution**, and justify the observed performance differences. **(15 Marks)**

iii. Create a comprehensive report recommending strategies to improve DNS performance, availability, and resilience for globally distributed services. **(10 Marks)**

4. Design and Develop a DNS Name Server Analysis Framework to Study Top-Level Domain Infrastructure (K6, CO6, WK3, WK4, WK5, WK8, WP1, WP2, WP3, EA5, SDG 9.4 & 12.5)

An Internet governance organization plans to evaluate the robustness of DNS infrastructures maintained by different Top-Level Domains (TLDs). As a network engineer, **design and develop** a framework to investigate the distribution of name servers and assess their impact on Internet scalability and reliability.

Tasks

i. Design and implement a methodology using the **dig +trace** command to identify and document the name servers associated with Top-Level Domains such as **.com, .in, .sg, .edu, and .org**. **(15 Marks)**

ii. Develop a comparative analysis of the number and distribution of name servers for selected TLDs by relating them to country size, Internet usage, and infrastructure development. Justify the observed trends using collected data. **(15 Marks)**

iii. Create a technical report recommending improvements to enhance DNS scalability, redundancy, and availability for TLD infrastructures. **(10 Marks)**

5. Design and Develop a DNS Security Assessment Framework for Enterprise Networks (K6, CO6, WK3, WK4, WK5, WK8, WP1, WP2, WP3, EA5, SDG 9.4 & 12.5)

A financial organization plans to strengthen the security of its DNS infrastructure against cyber threats such as spoofing, cache poisoning, and unauthorized DNS modifications. As a network security engineer, **design and develop** a DNS security assessment framework to identify vulnerabilities and recommend appropriate mitigation strategies.

Tasks

i. Design and implement a methodology to examine DNS records of selected domains using **dig, dig +dnssec**, or equivalent tools to identify vulnerabilities such as **absence of DNSSEC, insecure zone transfers, open resolvers, or misconfigured DNS records**. **(15 Marks)**

ii. Develop a risk assessment by classifying the identified vulnerabilities according to their severity and evaluating the security implications of different DNS server configurations (primary, secondary, caching, and recursive servers). **(15 Marks)**

iii. Create a comprehensive security report proposing suitable mitigation strategies, including **DNSSEC implementation, secure server configuration, redundancy, access control, and**

monitoring mechanisms, to improve the security and reliability of enterprise DNS infrastructure. **(10 Marks)**

Evaluation Criteria:

S.No	Question	Marks	KL	CO	PI
1.	Identifying the DNS records using network commands.	15	K3	CO1	1.4.1, 5.1.1
2.	Analyzing DNS record based on performance metrics.	15	K4	CO6	4.1.3, 4.2.1, 5.2.2, 9.1.1, 13.4.1
3.	Create a technical report based on the design and analysis of the solution.	10	K6	CO6	7.1.1, 8.3.1, 11.2.1, 12.4.1

Prepared By	Reviewed By	Approved By
Course Coordinator	PAC Team	HOD